



EAST TEXAS A&M
— UNIVERSITY —

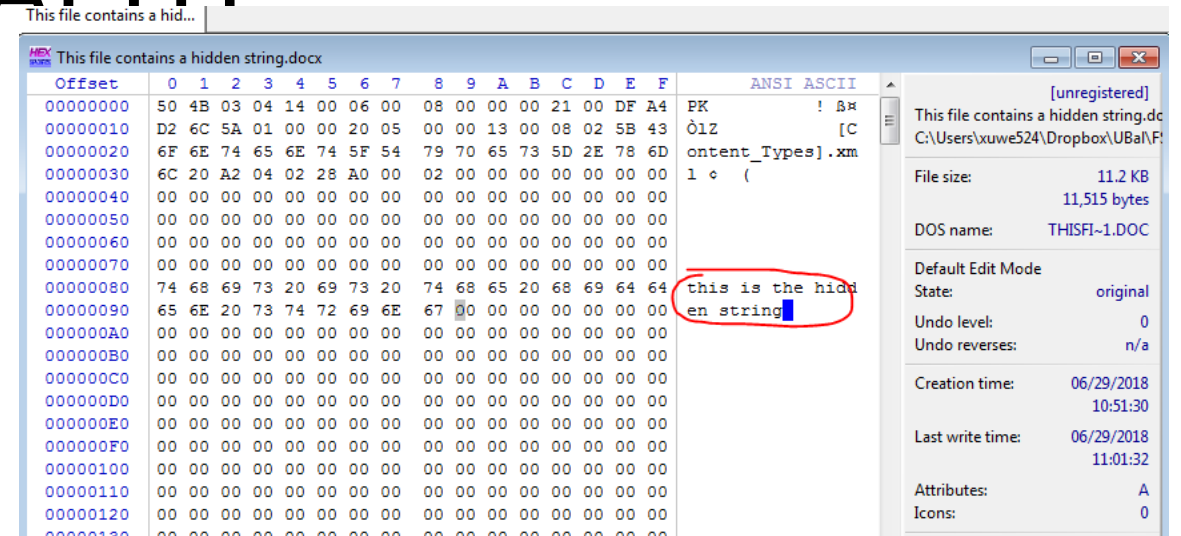
CSCI 352.61E

DIGITAL FORENSICS

STEGANOGRAPHY

WHAT IS STEGANOGRAPHY?

- Steganography hides information inside another file
 - Insertion
 - substitution: Replaces bits of the host file with other bits of data

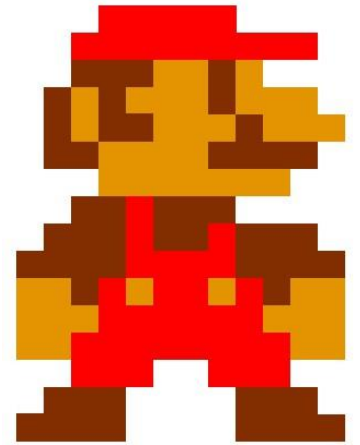
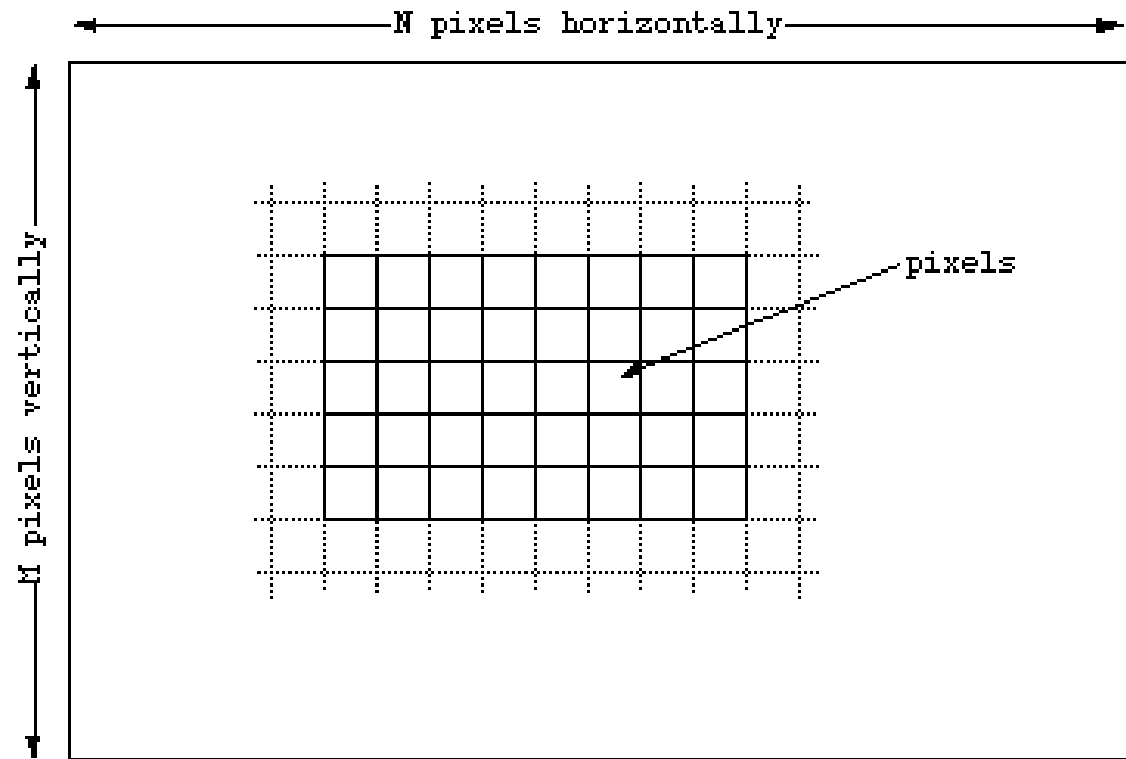


Original imageImage with a secret file

SUBSTITUTION

- Often use BMP(bitmaps) image
 - other formats are supported as well
- Replaces bits of the host file with other bits of data
- Usually change the last two LSBs (least significant bit)

BITMAPS



24 bit RGB



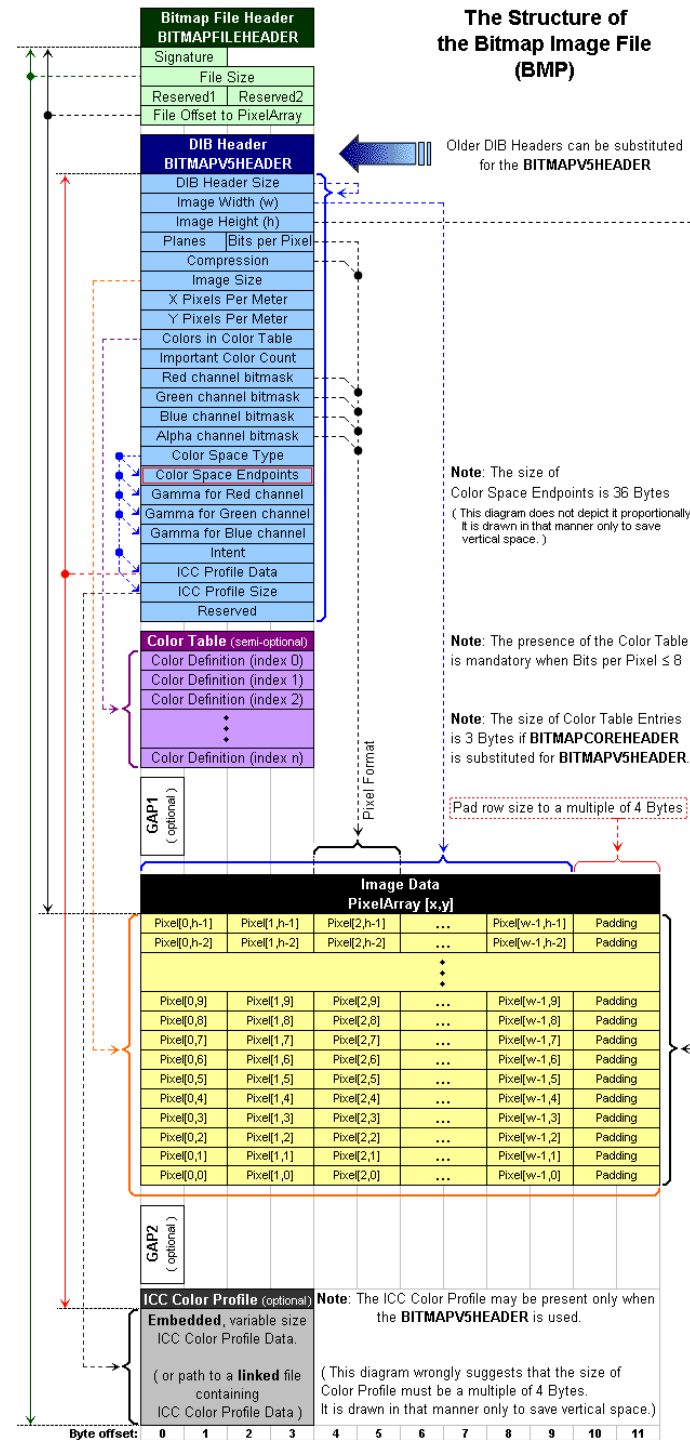
RGB Hex Triplet Color Chart

If you find this chart useful, please send Doug an e-mail note and say "Thanks!".
E-mail-Ware...What a concept! <http://home.earthlink.net/~jakesan/> jakesan@earthlink.net

FFFFFF	FFCCFF	FF99FF	FF66FF	FF33FF	FF00FF	NON-DITHERING COLORS
FFFFCC	FFCCCC	FF99CC	FF66CC	FF33CC	FF00CC	
FFFF99	FFCC99	FF9999	FF6699	FF3399	FF0099	
FFFF66	FFCC66	FF9966	FF6666	FF3366	FF0066	00FF00
FFFF33	FFCC33	FF9933	FF6633	FF3333	FF0033	00EE00
FFFF00	FFCC00	FF9900	FF6600	FF3300	FF0000	000000
CCFFFF	CCCCFF	CC99FF	CC66FF	CC33FF	CC00FF	00CC00
CCFFCC	CCCCCC	CC99CC	CC66CC	CC33CC	CC00CC	008000
CCFF99	CCCC99	CC9999	CC6699	CC3399	CC0099	00A000
CCFF66	CCCC66	CC9966	CC6666	CC3366	CC0066	009900
CCFF33	CCCC33	CC9933	CC6633	CC3333	CC0033	008800
CCFF00	CCCC00	CC9900	CC6600	CC3300	CC0000	007700
99FFFF	99CCFF	9999FF	9966FF	9933FF	9900FF	006600
99FFCC	99CCCC	9999CC	9966CC	9933CC	9900CC	005500
99FF99	99CC99	999999	996699	993399	990099	004400
99FF66	99CC66	999966	996666	993366	990066	003300
99FF33	99CC33	999933	996633	993333	990033	002200
99FF00	99CC00	999900	996600	993300	990000	001100
FF0000	66FFFF	66CCFF	6699FF	6666FF	6633FF	0000FF
EE0000	66FFCC	66CCCC	6699CC	6666CC	6633CC	0000EE
000000	66FF99	66CC99	669999	666699	663399	000000
CC0000	66FF66	66CC66	669966	666666	663366	0000CC
800000	66FF33	66CC33	669933	666633	663333	000080
A00000	66FF00	66CC00	669900	666600	663300	0000AA
990000	33FFFF	33CCFF	3399FF	3366FF	3333FF	000099
880000	33FFCC	33CCCC	3399CC	3366CC	3333CC	000088
770000	33FF99	33CC99	339999	336699	333399	000077
660000	33FF66	33CC66	339966	336666	333366	000066
550000	33FF33	33CC33	339933	336633	333333	000055
440000	33FF00	33CC00	339900	336600	333300	000044
330000	00FFFF	00CCFF	0099FF	0066FF	0033FF	000033
220000	00FFCC	00CCCC	0099CC	0066CC	0033CC	000022
110000	00FF99	00CC99	009999	006699	003399	000011
00FF66	00CC66	009966	006666	003366	000066	
00FF33	00CC33	009933	006633	003333	000033	
00FF00	00CC00	009900	006600	003300	000000	

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The Structure of the Bitmap Image File (BMP)



Original Pixel Color		Bit breakdown of a secret Letter “ A=(01 00 00 01)₂ ”	
1010 10 10	AA	1010 10 01	A9
1001 11 01	9D	1001 11 00	9C
1111 00 00	F0	1111 00 00	F0
0011 11 11		0011 11 01	

color-hex.com

color-hex 

color-hex 

HIDE A DATA FILE USING
STEGHIDE

STEGHIDE

- Hides data file in some of the least significant bits of another file
 - bmp, jpeg, wav and au files,
- Support encryption algorithms

```
(rootkali80)-[~/steg]  
# steghide
```

```
steghide version 0.5.1
```

```
the first argument must be one of the following:
```

embed, --embed	embed data
extract, --extract	extract data
info, --info	display information about a cover- or stego-file
info <filename>	display information about <filename>
encinfo, --encinfo	display a list of supported encryption algorithms
version, --version	display version information
license, --license	display steghide's license
help, --help	display this usage information

PREPARE LAB FILES

Download a bmp picture

```
(root@kali80)~[/steg]
# wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/steganography/_tower_original_image_for_lab.bmp

(root@kali80)~[/steg]
# md5sum _tower_original_image_for_lab.bmp
7f77e02253d5ae6426c4d0c932325ae3 _tower_original_image_for_lab.bmp
```

Create a file

```
(root@kali80)~[/steg]
# echo this is a secret file you want to hide! > secret.txt

(root@kali80)~[/steg]
# cat secret.txt
this is a secret file you want to hide!
```

```
(root@kali80)-[~/steg]  
# ls _tower original image for lab.bmp -l  
-rw-r--r-- 1 root root 9277494 Oct 14 22:11 _tower_original_image_for_lab.bmp
```

```
(root@kali80)-[~/steg]  
# display -resize 400x600 _tower original image for lab.bmp
```



HIDE A FILE

```
(rootkali80)-[~/steg]
# steghide embed -ef secret.txt -cf _tower original image for lab.bmp -p 1234
embedding "secret.txt" in "_tower_original_image_for_lab.bmp"... done

(rootkali80)-[~/steg]
# md5sum _tower original image for lab.bmp
0dcd7f484462e5e40887030c6808b074 _tower_original_image_for_lab.bmp

(rootkali80)-[~/steg]
# ls _tower original image for lab.bmp -l
-rw-r--r-- 1 root root 9277494 Oct 14 22:07 _tower_original_image_for_lab.bmp
```


EXTRACT A FILE

```
(root👤kali80)-[~/steg]  
# steghide extract -sf tower original image for lab.bmp -xf xsecret.txt -p 1234  
wrote extracted data to "xsecret.txt".
```

```
(root👤kali80)-[~/steg]  
# cat xsecret.txt  
this is a secret file you want to hide!
```

CRACKER PASSWORD

Find a password dictionary

```
(root👁kali80)-[~/steg]
# ls /usr/share/wordlists
dirb          dnsmap.txt    fern-wifi     nmap.lst      seclists
dirbuster     fasttrack.txt metasploit    rockyou.txt.gz wfuzz

(root👁kali80)-[~/steg]
# gunzip /usr/share/wordlists/rockyou.txt.gz

(root👁kali80)-[~/steg]
# ls /usr/share/wordlists/rockyou.txt
/usr/share/wordlists/rockyou.txt
```

```
(rootkali80)-[~/steg]
```

```
# stegcracker _tower_original_image_for_lab.bmp /usr/share/wordlists/rockyou.txt
```

```
StegCracker 2.1.0 - (https://github.com/Paradoxis/StegCracker)
```

```
Copyright (c) 2021 - Luke Paris (Paradoxis)
```

StegCracker has been retired following the release of StegSeek, which will blast through the rockyou.txt wordlist within 1.9 second as opposed to StegCracker which takes ~5 hours.

StegSeek can be found at: <https://github.com/RickdeJager/stegseek>

```
Counting lines in wordlist..
```

```
Attacking file '_tower_original_image_for_lab.bmp' with wordlist '/usr/share/wordlists/rockyou.txt'..
```

```
Successfully cracked file with password: 1234
```

```
Tried 2037 passwords
```

```
Your file has been written to: _tower_original_image_for_lab.bmp.out
```

```
1234 ←
```



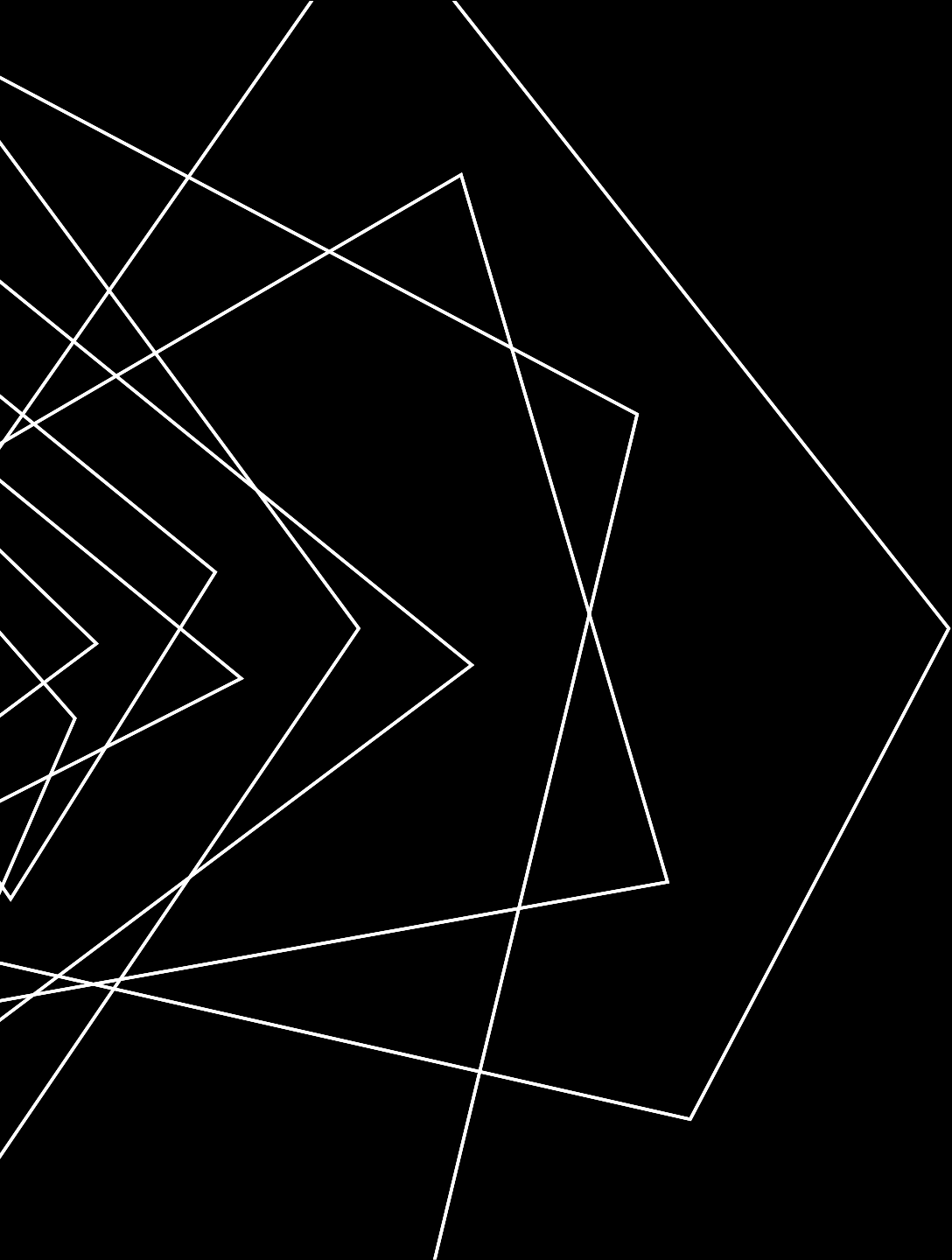
```
yoga@csci:~/csci352/A3$ wget -q https://github.com/danielmiessler/SecLists/raw/master/Passwords/Leaked-Databases/rockyou.txt.tar.gz
yoga@csci:~/csci352/A3$ ls
rockyou.txt.tar.gz  secret.txt  _tower_original_image_for_lab.bmp  xsecret.txt
yoga@csci:~/csci352/A3$ tar -xvzf rockyou.txt.tar.gz
rockyou.txt
yoga@csci:~/csci352/A3$ ls
rockyou.txt  rockyou.txt.tar.gz  secret.txt  _tower_original_image_for_lab.bmp  xsecret.txt
yoga@csci:~/csci352/A3$ stegseek _tower_original_image_for_lab.bmp rockyou.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "9876"
[i] Original filename: "secret.txt".
[i] Extracting to "_tower_original_image_for_lab.bmp.out".
```

FASTER CRACKING (STEGSEEK)

<https://github.com/RickdeJager/stegseek>

password	Line	Stegseek v0.6	Stegcracker 2.0.9	Stegbrute v0.1.1 (-t 8)
"cassandra"	1 000	0.05s	3.1s	0.7s
"kupal"	10 000	0.05s	14.4s	7.1s
"sagar"	100 000	0.09s	2m23.0s	1m21.9s
"budakid1"	1 000 000	0.73s	[p] 23m50.0s	13m45.7s
"_____1"	14 344 383	1.21s	[p] 5h41m52.5s	[p] 3h17m38.0s



THANK YOU

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